

Mid-term examination on Introduction to the Internet

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1. What advantage does a circuit-switched network have over a packet-switched network? What advantage does TDM have over FDM in a circuit-switched network?
2. Why is it that packet switching is said to employ statistical multiplexing? Contrast statistical multiplexing with the multiplexing that takes place in TDM.
3. What is meant by connection state information in a virtual circuit network?
4. Suppose you are developing a standard for a new type of network. You need to decide whether your network will use virtual circuits (VC) or datagram routing. What are the pros and cons for using VCs?
5. What are the advantages of message segmentation in packet-switched networks? What are the disadvantages?
6. What is the key distinguishing difference between a tier-1 ISP (backbone) and a tier-2 ISP?
7. Is HFC bandwidth dedicated or shared among users? Are collisions possible in a downstream HFC channel?
8. Consider sending a series of packets from a sending host to a receiving host over a fixed route. List the delay components in the end-to-end delay for a single packet. Which of these delays are constant and which are variable?
9. List five tasks that a protocol layer can perform. Is it possible that one (or more) of these tasks could be performed by two (or more) layers?